Problem description:

- The problem focuses on controlling a traffic signal at an intersection
- At an intersection, cars arrive from two directions, north-south (NS) and east-west (EW)
- The traffic signal can allow either ns to move(green) or ew to move while the other direction waits.
- The objective is to reduce the number of waiting vehicles at the intersection.
- This is modelled as MDP where state represents number of cars waiting in each direction and current signal status.
- The agent (traffic controller) decides about the signal switching.
- A reward function is defined based on total no of waiting cars, more traffic leads to negative reward.
- Value iteration algorithm is used
- We use parameters such as gamma (discount factor) and theta (convergence threshold)

1. Discount factor:

High values of gamma (0.75-0.9): make the agent focus more on long-term-traffic low

2. influence of theta:

Small theta 1e-6:

- More precise and convergence
- More iterations required
- Slightly higher than computation time

Larger theta:

- Policy obtained with theta = 1e-2 and 1e-6 was often similar, indicated very strict convergence
- In some cases, theta = 1e-3 produced better results because algorithm converged faster and avoided over-refining value estimates.

- So, theta = 1e-3 has good-trade-off between policy and computational efficiency

3. traffic queue behaviour

- Average ns queue length range is between 3.6 and 11 cars
- Average EW queue length ranged between 5 and 13 cars
- Better policies showed balanced waiting vehicles and lower maximum queue length
- When queue was balanced (6-7 cars) congestion events less traffic.

4. congestion events

- This is length exceeding 10 cars
- Poor policies predicted 70-88 congestion events

- Better policies produced 15-16 events.

5. traffic light switching behaviour:

- Higher switching showed system reacting to changing traffic
- Best policies showed moderate switching (65-69)

6. overall performance

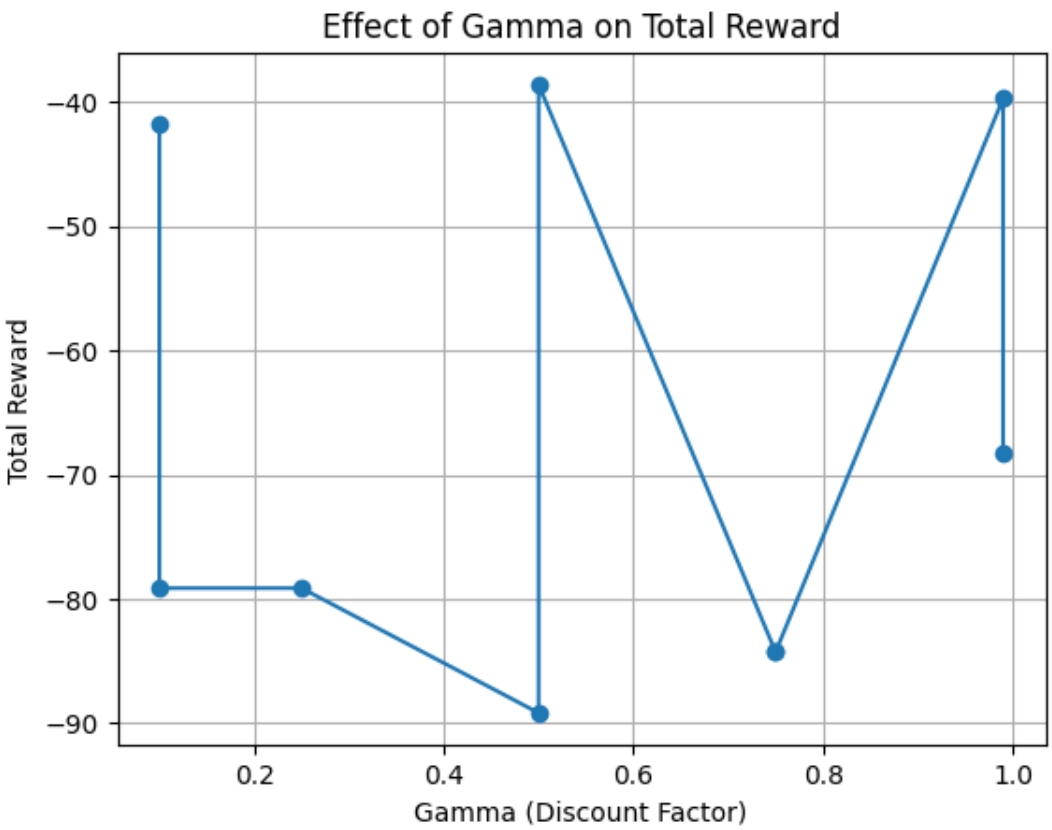
- Best policies achieved
- Higher reward, lower congestion event
- Balanced queue in both sides

Gamma (γ)	Theta (θ)	Total Reward	Avg NS Cars	Avg EW Cars	Max NS Queue	Max EW Queue	Congestion Events	Light Switches	Performance
0.99	1e-6	-68.33	4.47	13.03	15	20	72	56	Poor
0.99	1e-3	**-39.73**	**3.62**	7.70	12	20	26	64	Good
0.75	1e-6	-84.27	10.29	12.59	17	20	83	60	Very Poor
0.75	1e-3	-84.27	10.29	12.59	17	20	83	60	Very Poor
0.50	1e-1	**-38.60**	6.45	**5.13**	18	14	**15**	67	**Best**
0.50	1e-3	-89.17	11.19	12.56	19	20	88	60	Worst
0.25	1e-3	-79.13	10.94	10.40	20	20	73	64	Poor
0.10	1e-6	-79.13	10.74	10.60	20	20	70	69	Poor
0.10	1e-3	-41.77	6.99	5.54	15	14	19	69	Good

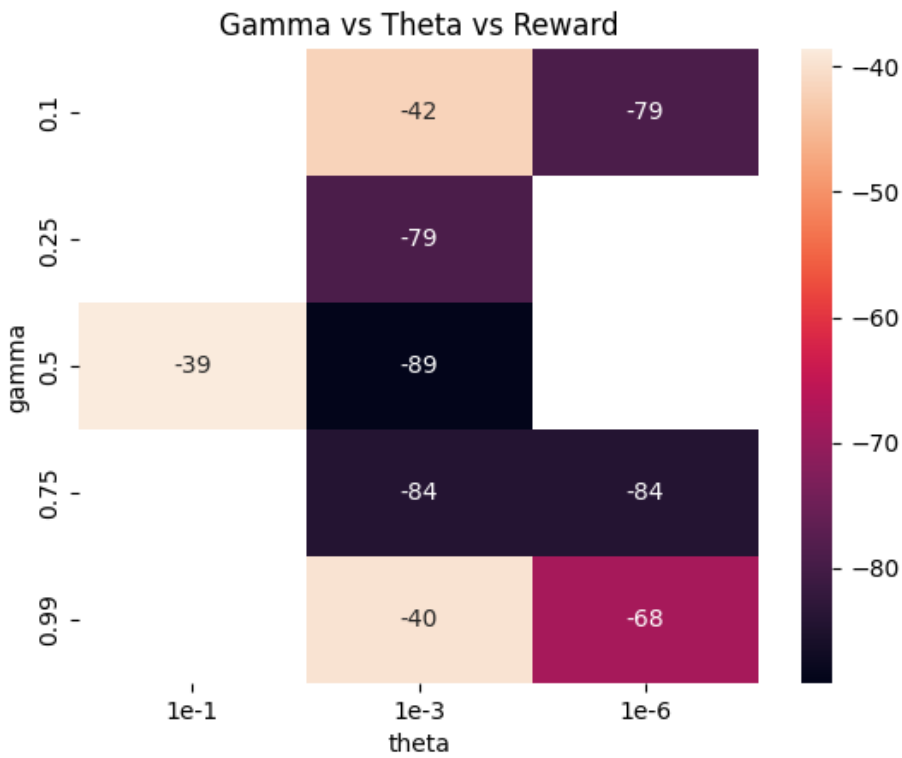
The best performance was with gamma = 0.50 and theta = 1e-1

Visualization:

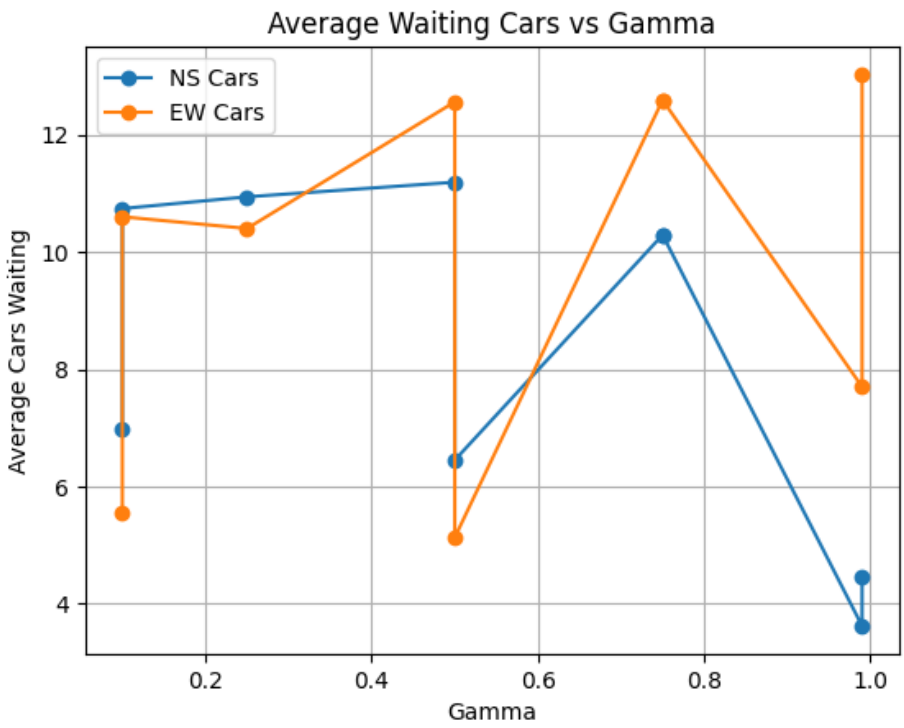
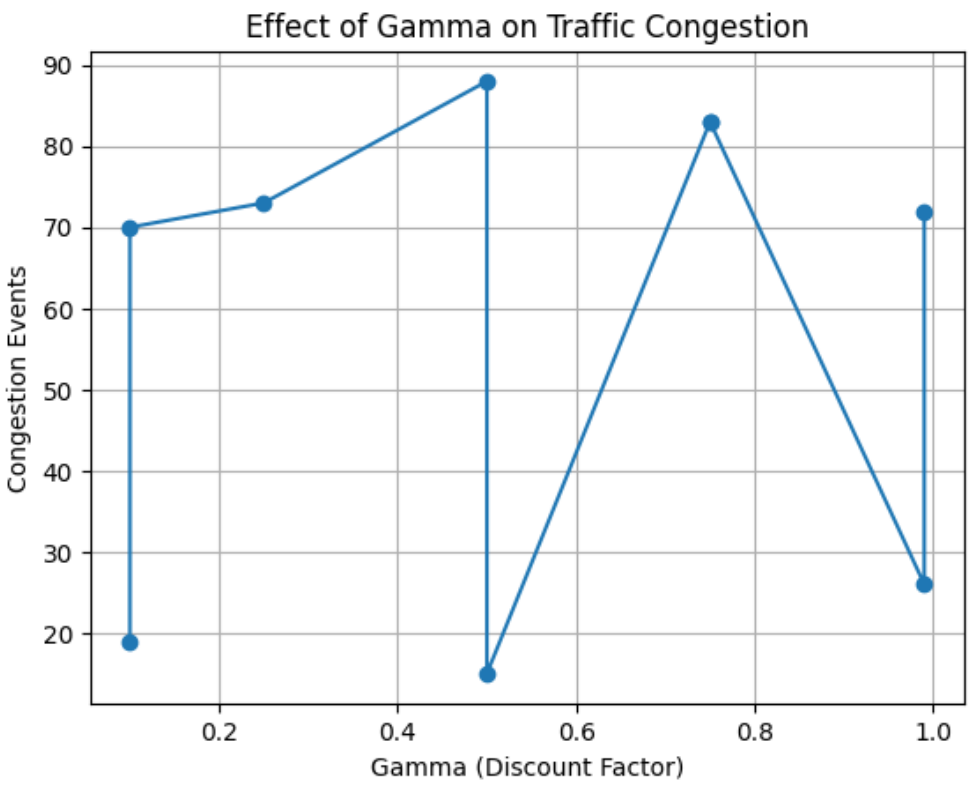
1. Moderate gamma values produced better rewards, indicating a balanced consideration of immediate and future traffic conditions.



2. Lower and moderate gamma values resulted in fewer congestion events compared to higher gamma values



3. Balanced average waiting vehicles in NS and EW directions indicate more efficient traffic signal decisions.



4. The heatmap shows that certain combinations of gamma and theta lead to better rewards, highlighting the importance of tuning both parameters.